

Control Unit Architecture & Hardwired/Microprogrammed Control

Q. What is the role of the Control Unit? Differentiate between Hardwired Control Unit and Microprogrammed Control Unit.

> **Definition to Remember**

1. Hardwired Control Unit

The control logic is implemented physically using digital hardware circuits (logic gates, flip-flops, decoders).

2. Microprogrammed (Soft-Wired) Control Unit

The control signals are generated by executing a tiny, low-level program (a **Microprogram**) stored in a special, fast memory called **Control Memory (ROM)**.

3. Key Differences

Feature	Hardwired CU	Microprogrammed CU
Control Logic	Implemented physically using digital hardware circuits.	Generated by executing a microprogram stored in Control Memory (ROM).
Flexibility	Not flexible; changes require physical redesign.	Flexible; changes can be made by updating the microprogram.
Cost	Higher cost due to complex hardware.	Lower cost due to simpler hardware and software-based control.
Development Time	Longer development time for physical redesign.	Shorter development time for software updates.

> **Must-Write Points (for 10 marks)**

> **Quick Recall**

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